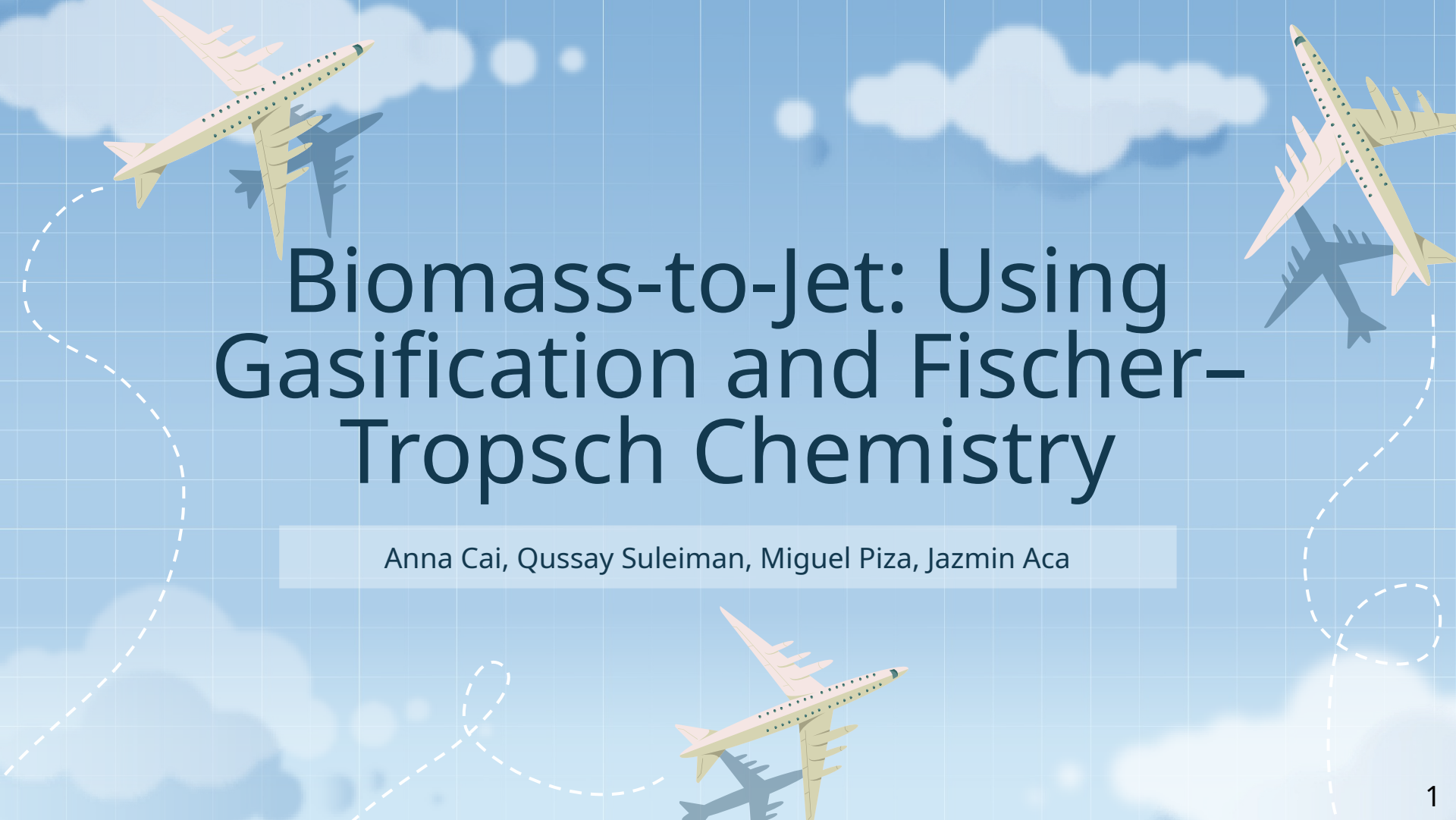
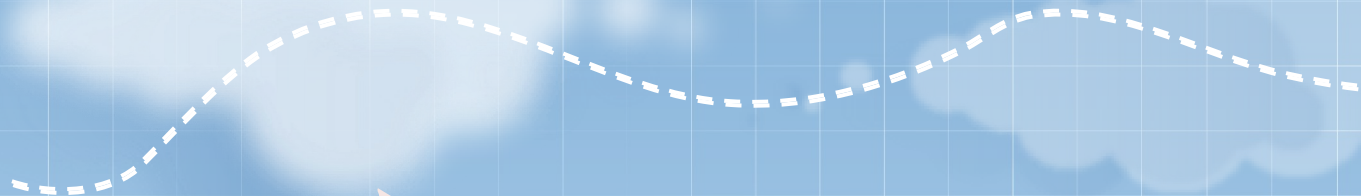




Biomass-to-Jet: Using Gasification and Fischer-Tropsch Chemistry

Anna Cai, Qussay Suleiman, Miguel Piza, Jazmin Aca



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introduction

What is saf and why

- Renewable jet fuel derived from **sustainable feedstocks**
- Designed as a “drop-in” replacement for conventional Jet A
- Fischer-Tropsch Synthetic Paraffinic Kerosene (FT-SPK) has an up to 50% blending potential

What we do:

- Design a process to convert wood-chip-based biomass into syngas
- Clean and feed syngas to a Fischer-Tropsch reactor to derive useful hydrocarbons
- Send hydrocarbons to upgrading processes to isolate products



Net-zero emissions goal by 2050

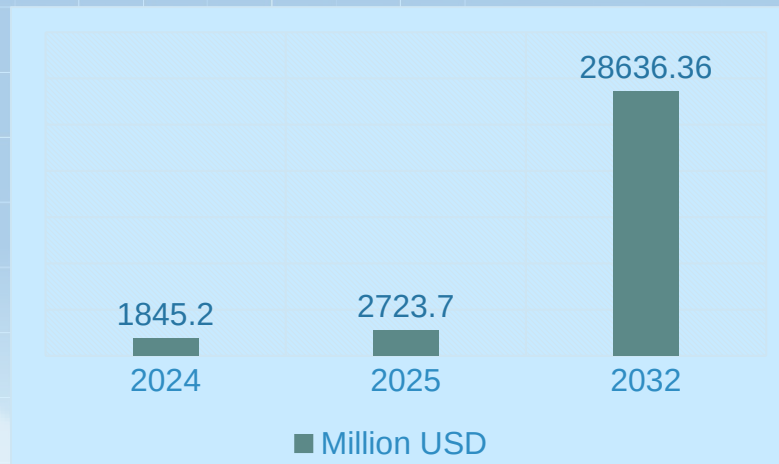


The **SAF Grand Challenge** aims to scale the production of **SAF** in the U.S. to reduce greenhouse gas emissions from aviation

Market size and Forecast

Sustainable Aviation Fuel Market to grow at **39.95% CAGR** during 2025–2032

Driven by rising demand for alternative fuels, new technologies, SAF mandates, and sustainability initiatives



North American
market share

\$569.5 Million

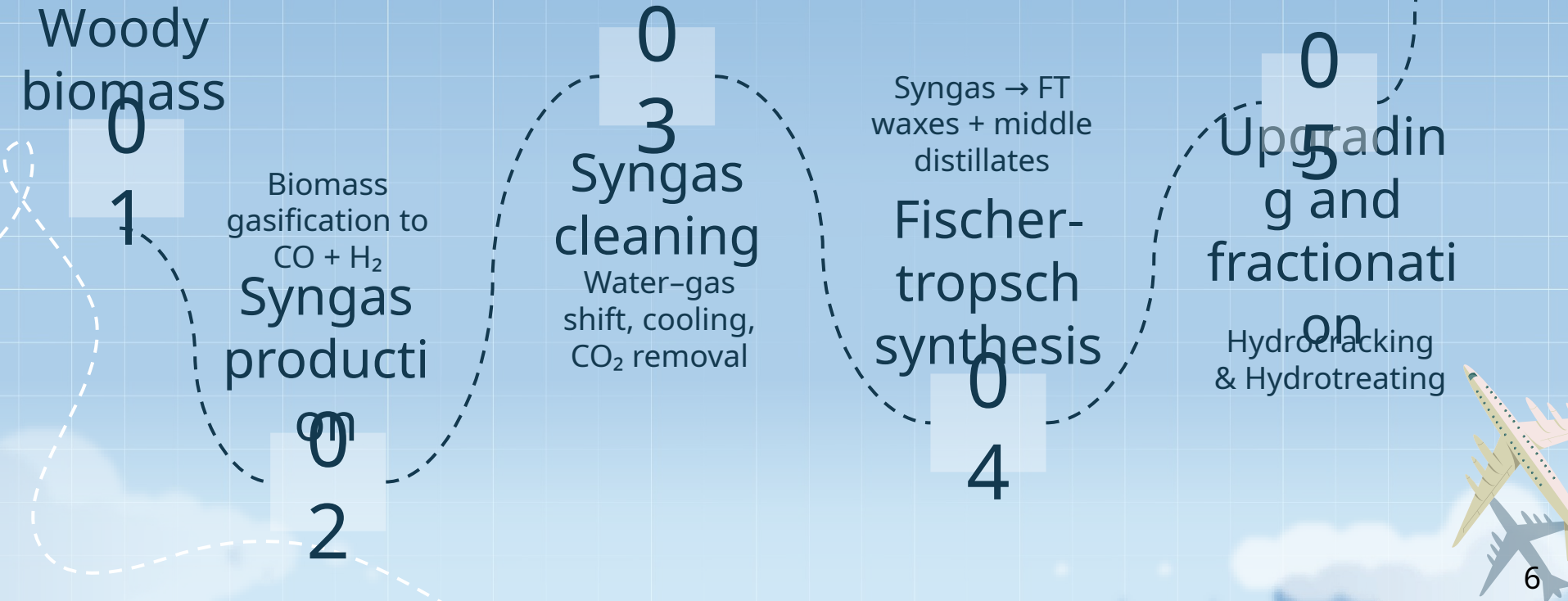
2023

\$848.8 million

2024



Project overview



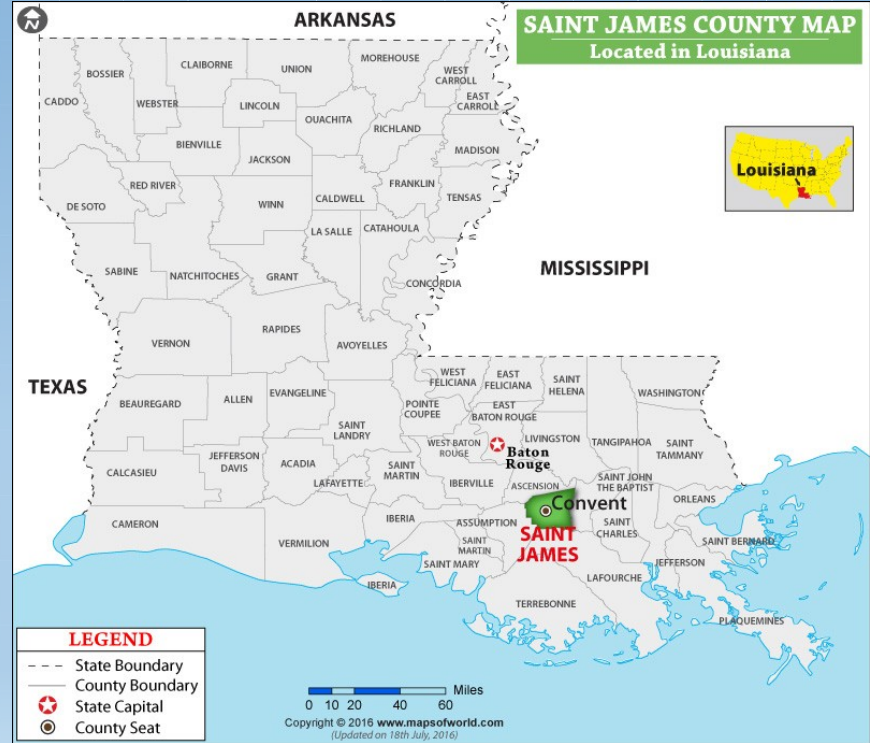
Design basis

Plant Capacity

- 10,000 barrels per day of SAF
- 3,500 barrels per day of Naphtha

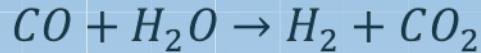
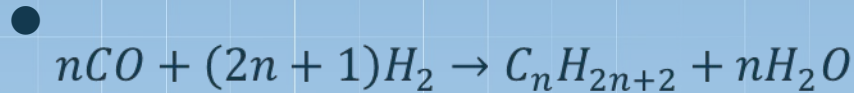
Location: Saint James County, Louisiana (U.S. Gulf Coast)

- Abundant forestry and agricultural biomass
- Strong hydrogen availability
- Deep-water port access (New Orleans, Baton Rouge)
- High SAF demand from nearby major airports



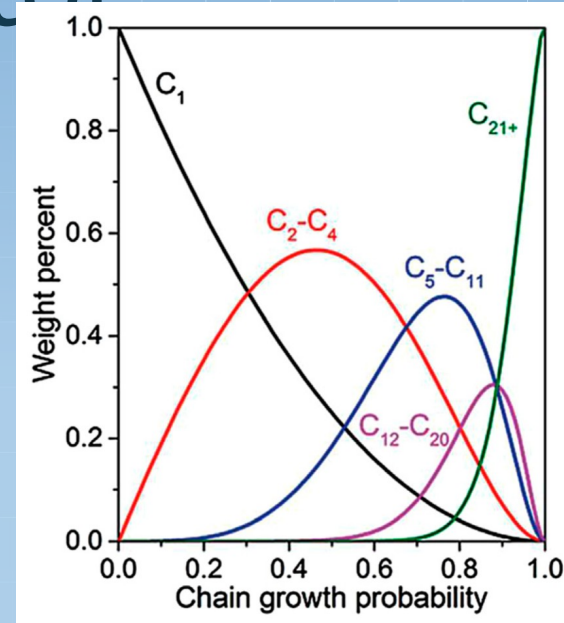
Malik, V. *St James Parish Map, Louisiana*

Fischer-tropsch reactor

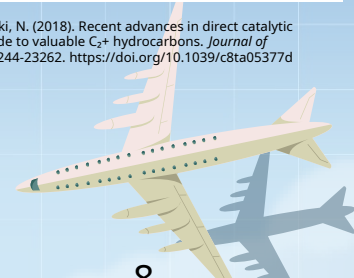


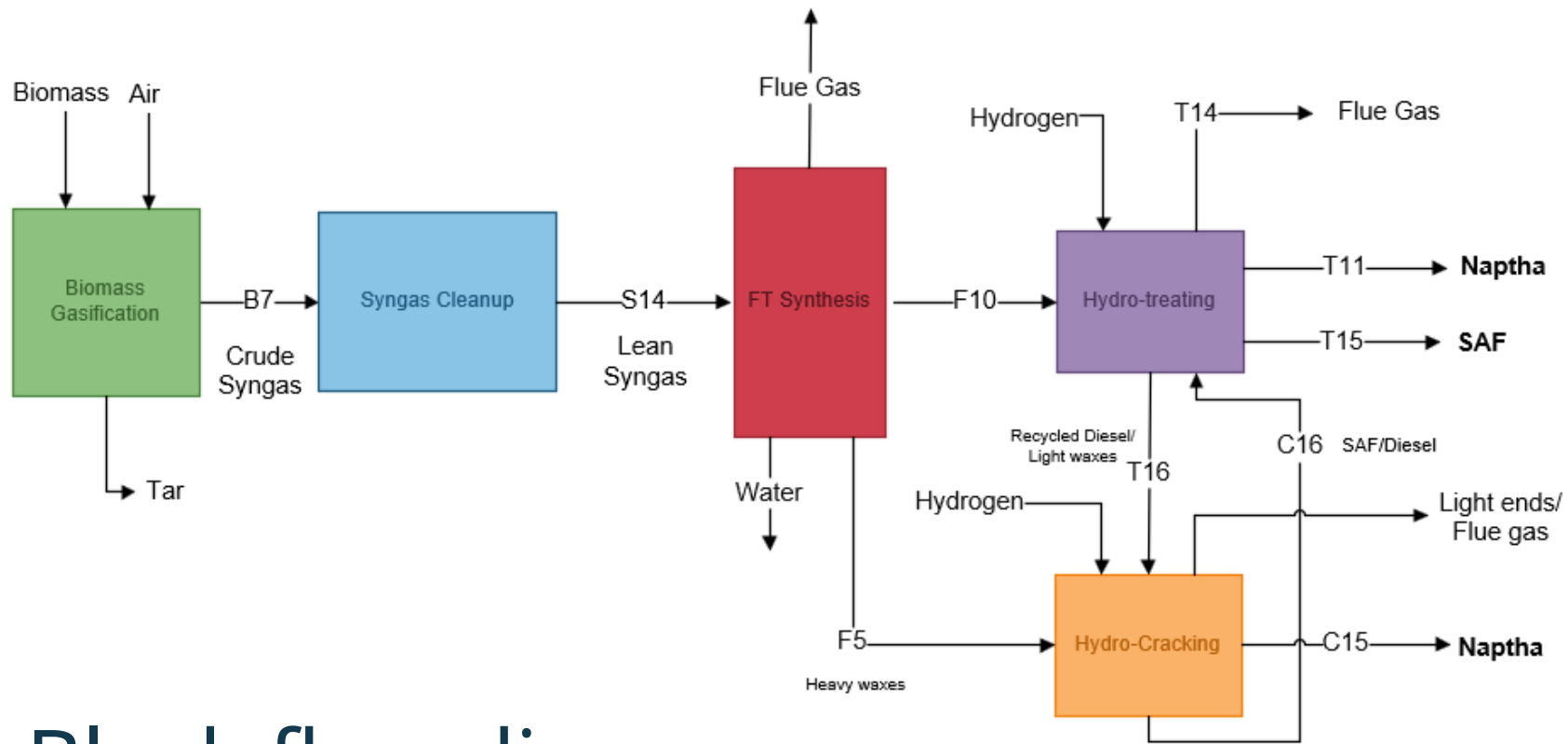
Molecular weight distribution according to
Anderson-Schulz-Flory distribution

$$\frac{W_n}{n} = 1 - \alpha^2 \alpha^{n-1}$$



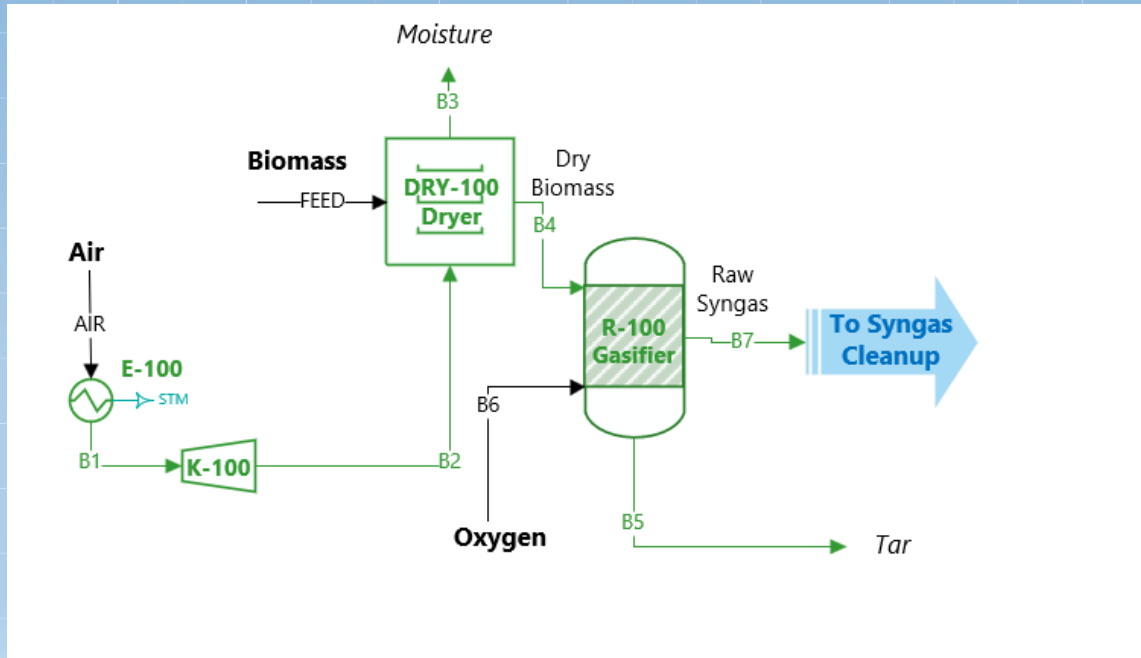
Guo, L., Sun, J., Ge, Q., & Tsubaki, N. (2018). Recent advances in direct catalytic hydrogenation of carbon dioxide to valuable C_2+ hydrocarbons. *Journal of Materials Chemistry A*, 6(44), 23244-23262. <https://doi.org/10.1039/c8ta05377d>



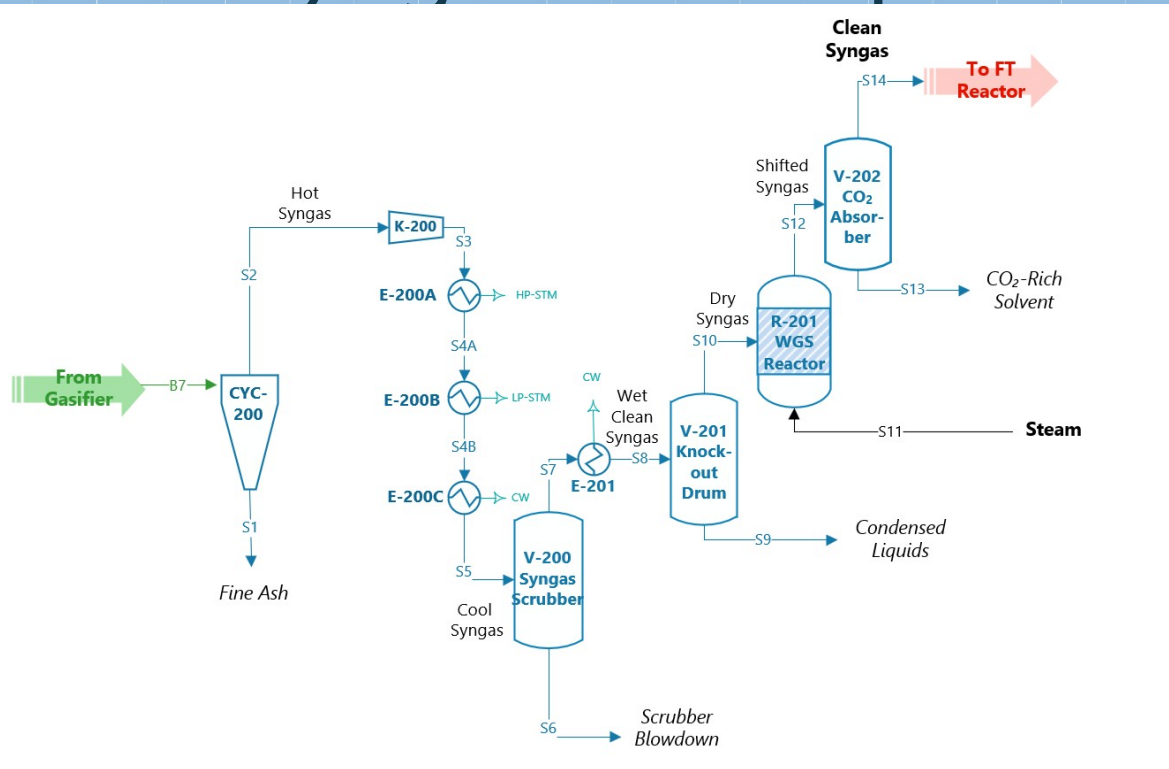


Block flow diagram

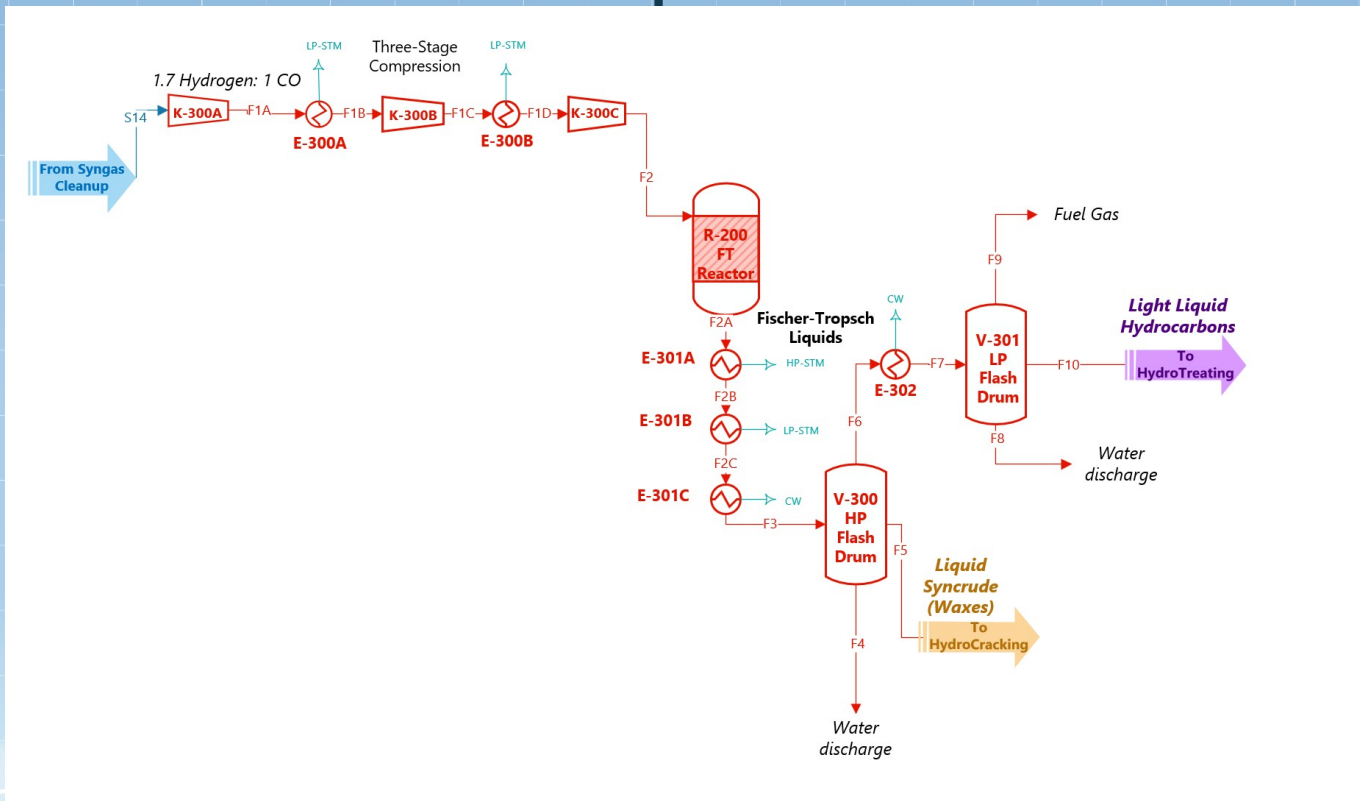
Biomass Gasification



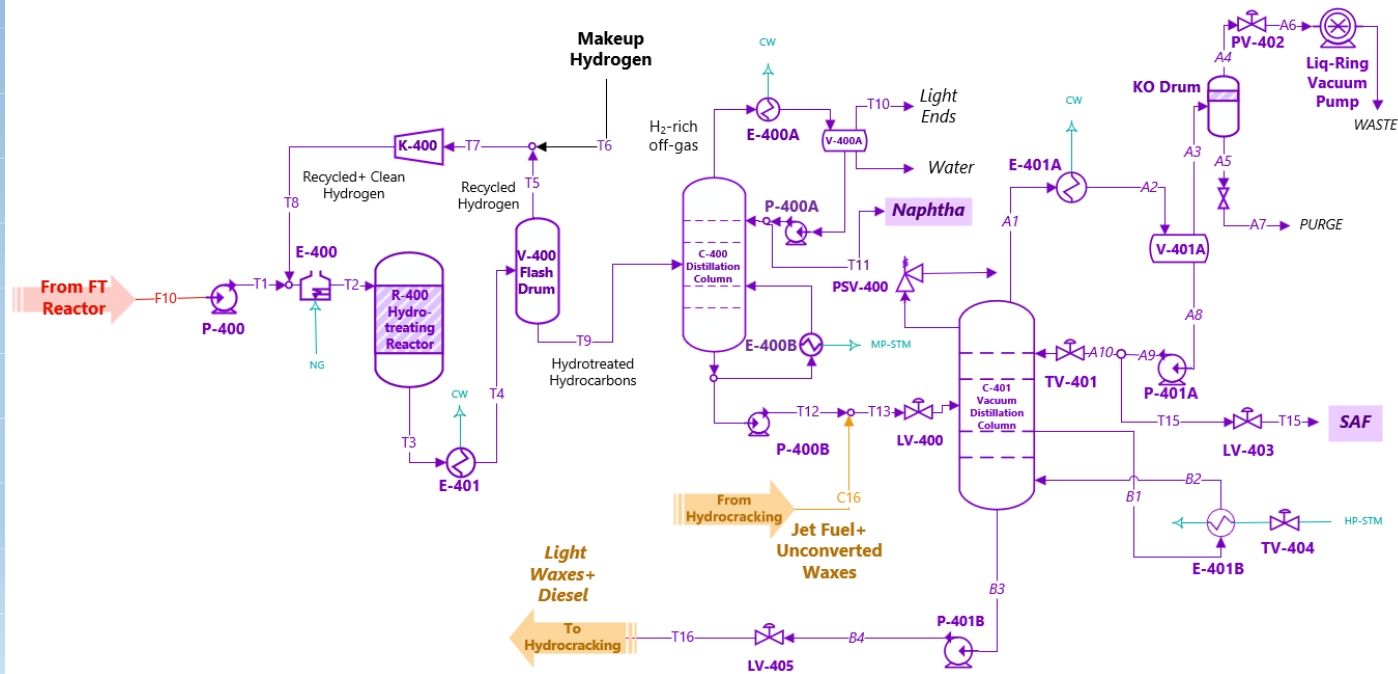
Syngas cleanup



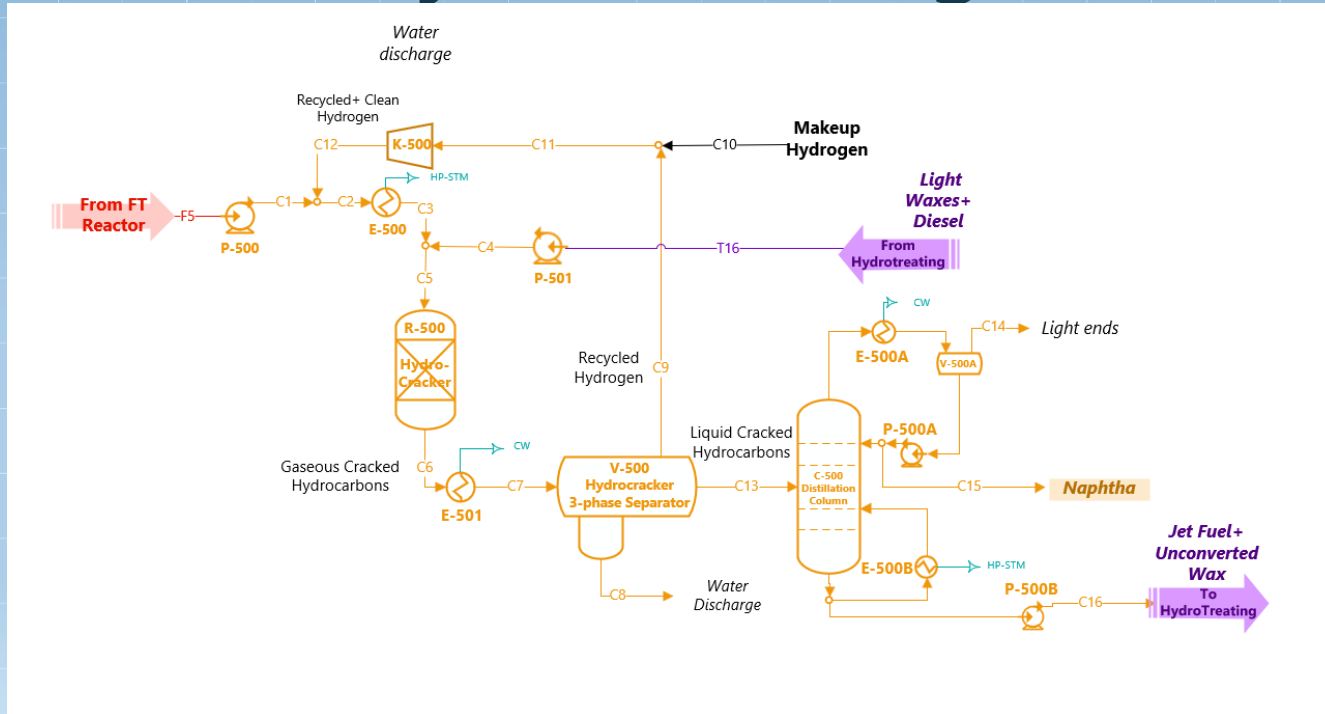
Fischer tropsch reactor



hydrotreating



hydrocracking





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safety

Operational safety concerns

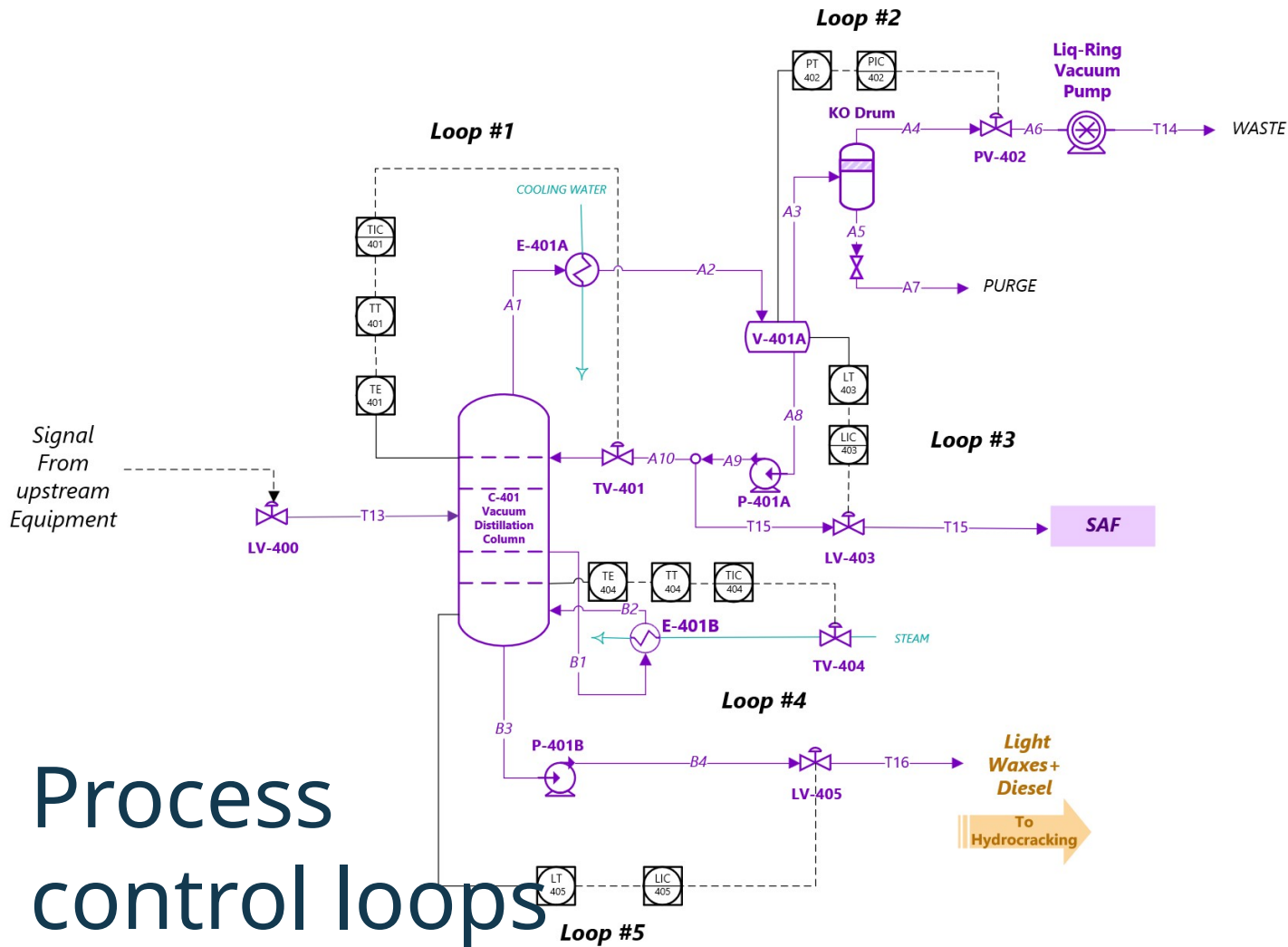
Unit	Safety Concerns	Prevention and Mitigation
Biomass Dryer	Fire or Explosion Conditions	Temperature monitoring and high temperature shutdown.
	Dust from woodchips	Install dust suppression, proper ventilation, and enforce proper respiration protection
Biomass Gasifier	High Temperatures	Temperature monitoring and high temperature shutdown.
	Flammable gasses	Gas leak detection
Water gas shift reactor	Temperature rise	Tight temperature monitoring across bed
	Overpressure	Pressure relief valves
	CO Leaks	Use CO detection around reactor area
CO₂ Absorber	CO ₂ -rich solvent leaks	Provide secondary containment for possible solvent leaks
	Column Flooding	Differential Pressure monitoring
Fischer-Tropsch Reactor	Highly Exothermic Reaction	Temperature measurements across reactor
	CO and H ₂ Leaks	Feed control, pressure relief to safe disposal system
	Runaway temperatures	High-high temperature emergency shutdown
Hydrotreating Reactor	High pressure hydrogen and hydrocarbon mixture	High-high temperature and pressure trips

Operational safety concerns

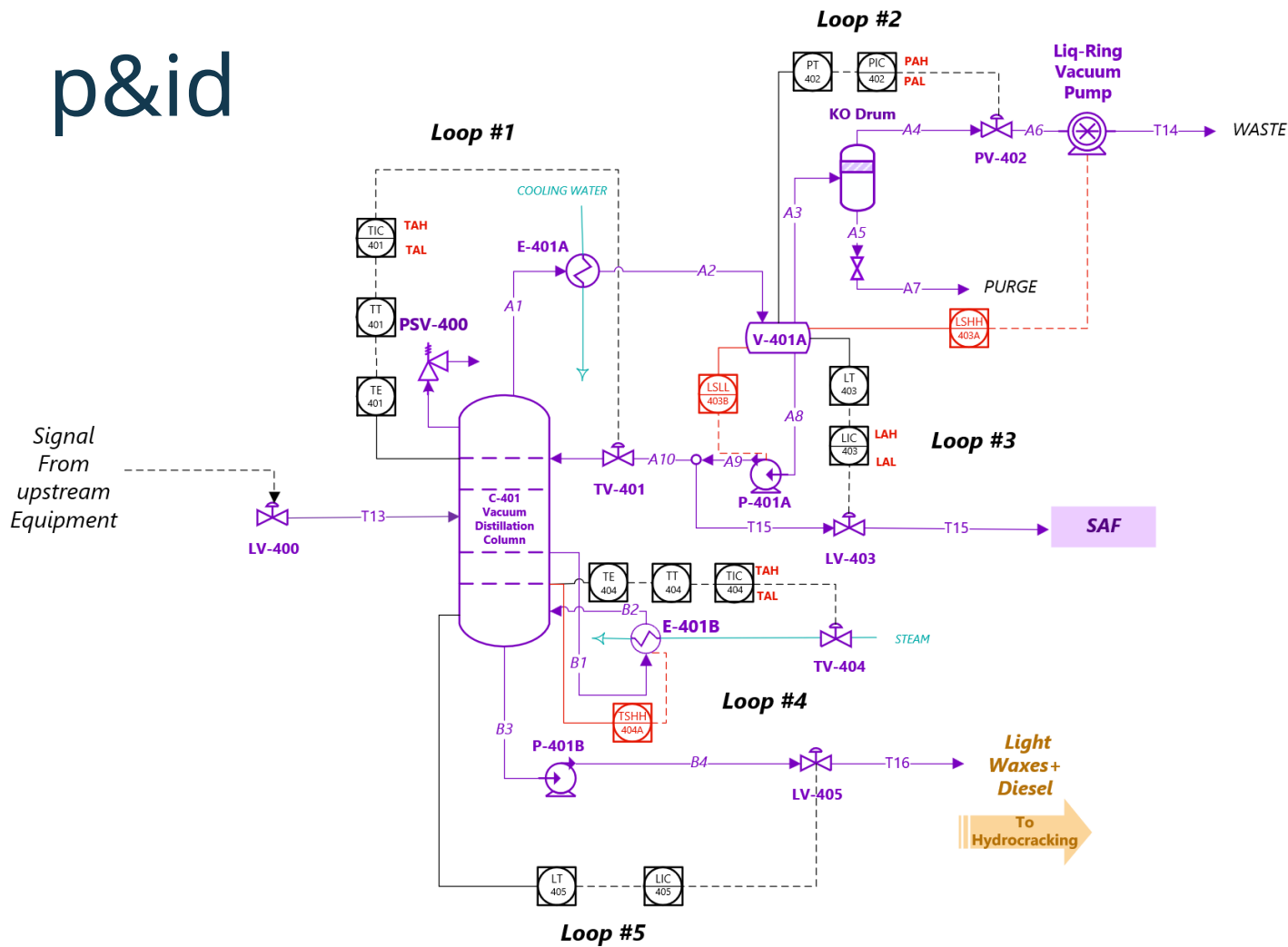
Unit	Safety Concerns	Prevention and Mitigation
Hydrotreater Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Flooding	DP monitoring
Hydrotreater Vacuum Distillation Column	Vacuum Loss	Vacuum monitoring and alarms
Hydrocracker	High pressure hydrogen	Route relief streams safely
	Strong exothermic reaction potential	Multiple temperature measurements
	hotspot formation	
	Hydrogen leak and fire risk	Leak detection and monitoring, flammable gas detectors
3-phase separator	Hydrogen rich gas release	Closed vent handling
Hydrotreater Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Over head vapor release	Safe overhead vent handling



Process control loops



p&id



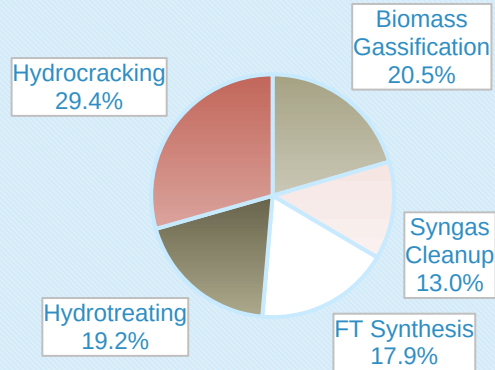


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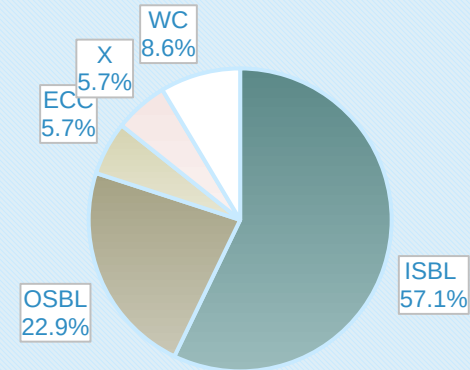
Process economics

Capital costs

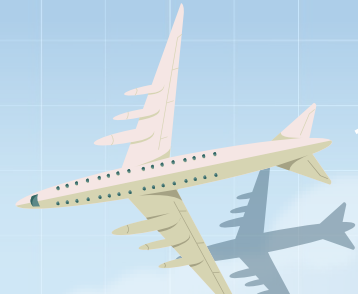
ISBL Breakdown



Capital Cost Breakdown

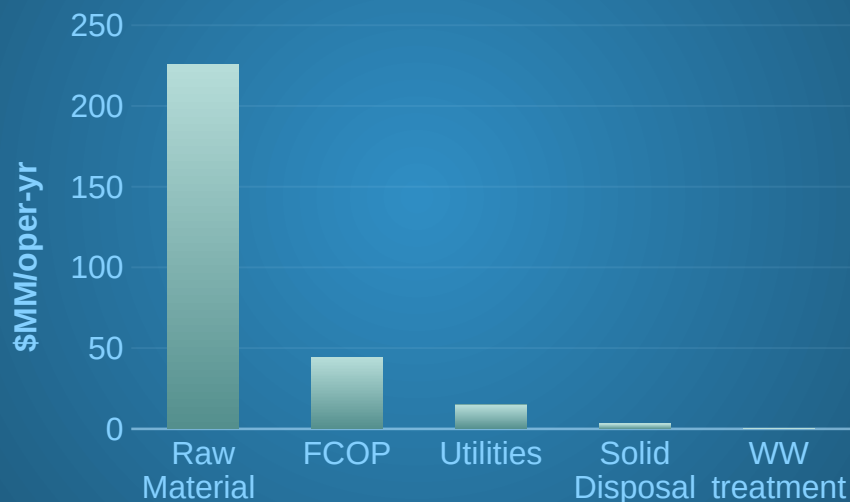


Type	ISBL	OSBL	ECC	X	WC	TCC
Cost (\$MM)	560.3	224.1	56.0	56.0	84.0	980.6

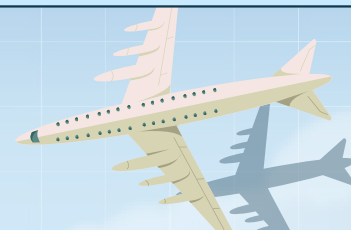


Variable Costs

CCOP Breakdown



Type	Cost (\$MM/oper-yr)
Raw Materials	225.8
Utilities	14.8
WW treatment	0.04
Solids Disposal	3.13
FCOP	44.3
CCOP	288.1

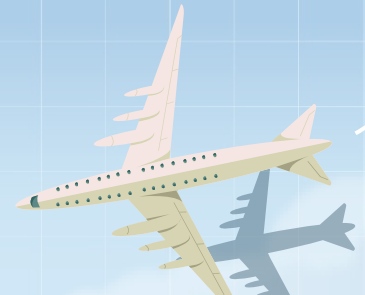


Revenue and gross profit

Product Type	Revenue (\$MM/yr)
SAF	353.6
Naphtha	77.16
Total	430.8

Credit Type	Revenue (\$MM/yr)
SAF	198.0
Naphtha	118.15
Total	316.2

Total Revenue	746.9 \$MM/yr
Gross Profit	458.8 \$MM/yr

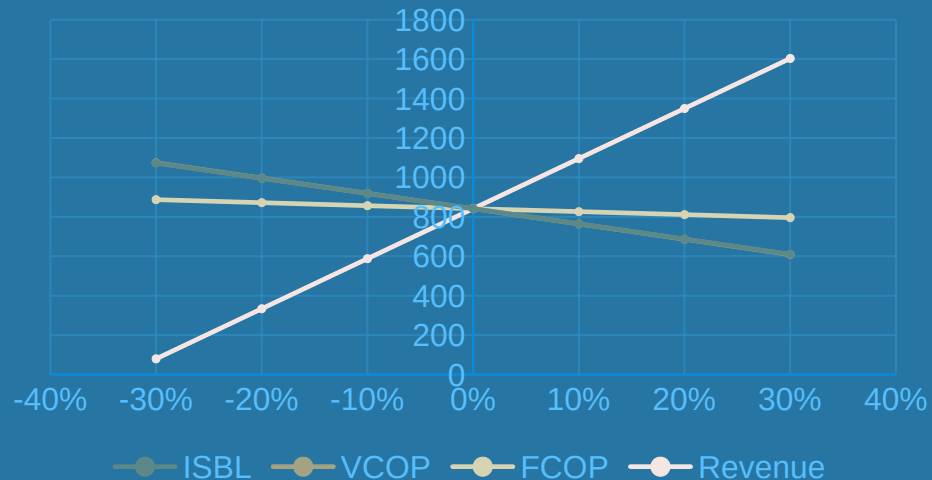


Economic Analysis

MACRS-7 yr NPV



Sensitivity Analysis

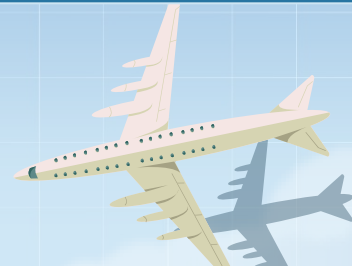


Constants

Tax Rate	26.5%
Interest Rate	15%

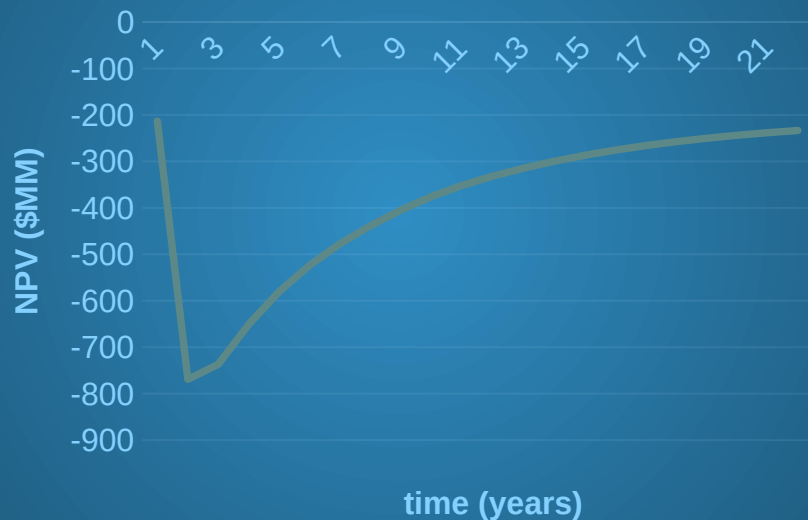
DCFOR (7-yr) 31.4%

DCFOR (5-yr) 31.6%



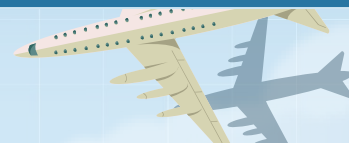
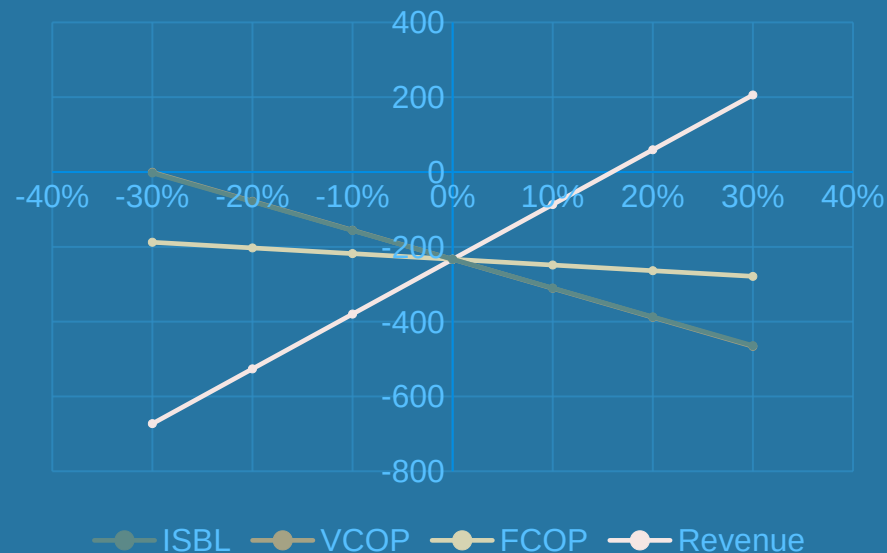
Economic Analysis-no

MACRS-7 yr NPV



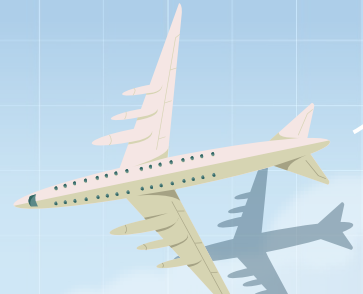
DCFOR (7-yr) 9.4%

Sensitivity Analysis



Further Recommendations

- Look to improve heat integration
- Consider incorporating Electrolyzation for Oxygen and Hydrogen
- Reduction of biomass feed even further via other methods of syngas creation (Carbon Capture and catalytic CO₂ reforming and partial oxidation reforming)
- Consider implementing catalyst regeneration
- Implement renewable energy



Thank You!

Any Questions?



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